Introduction

This report documents the iterative development, evaluation, and optimization of filter designs for the cochlear implant application.

The initial filter design adopted from Phase 2, referred to as Filter Design #1, served as the baseline for all iterations. Each subsequent design was derived from Filter Design #1, with a single parameter varied—such as adding overlapping subbands or modifying the filter type—while keeping all other parameters constant. This approach ensured that the impact of each parameter on the filter's performance could be isolated and thoroughly studied.

After evaluating all design iterations, a select few were identified as top-performing filters. These designs were further refined by varying the number of channels to determine the optimal configuration for the cochlear implant's use case.

Evaluation Method

The filter designs were evaluated using 16 sounds selected in Phase 1, covering three categories: speech, man-made, and environmental sounds. The evaluations were assessed using metrics defined in Phase 1: clarity, intelligibility, performance in multi-speaker scenarios, sound naturalness, similarity to input, and noise reduction performance. Each metric was weighted differently across the three sound categories based on its relevance, as summarized in Table 1.

Table 1: Weighted evaluation metrics across three sound categories.

Sound Categories	Subjective Metric Weights	Objective Metric Weights
Speech	Clarity: 30% Intelligibility: 20% Performance in Multi-Speaker Scenarios: 20%	Noise Reduction Performance: 15% Similarity to Input: 15%
Man-Made	Sound Naturalness: 30%	Noise Reduction Performance: 30% Similarity to Input: 40%
Environment	Sound Naturalness: 50%	Similarity to Input: 50%

In Phase 1, scoring systems varied, with some metrics rated on a 1-5 scale and others as a percentage. To simplify calculations, all scoring systems were standardized to a 0-100 scale for this phase.

For evaluations, all 16 sounds were processed through the filter design and played to four listeners. Each listener assessed the filtered sounds using the metrics on the new standardized scale. Scores for each sound metric were first averaged among the listeners, then grouped into category averages, and ultimately combined into an overall average for the filter design, as shown in Table 2. For the full evaluation scores, refer to the Appendix.

Table 2: Summary of filter design average scores.

Filter Design	Speech	Man-made	Environmental	Overall Average
1	26.41	35.98	43.13	32.09
2	35.22	33.40	53.44	36.81
3	21.43	22.48	36.25	23.68
4	20.07	25.33	38.75	24.38
5	40.52	44.79	45.31	42.72

Results of Filter Design #1

Filter Design #1 served as the baseline filter for all subsequent iterations. It uses a 17-band Butterworth filter design (order 2) with logarithmic spacing across the frequency range of 100 Hz to 8 kHz. As shown in Table 2, this design achieves an overall average score of 32.09, with individual category scores of 26.41 for speech, 35.98 for man-made sounds, and 43.13 for environmental sounds.

While this filter performed moderately well across all categories, it struggled to achieve high scores in clarity and intelligibility for speech sounds, as noted in the Appendix.

This filter achieved moderate scores across all evaluation metrics. Listeners reported that the audio samples sounded distorted and overly processed, with speech being barely intelligible and difficulty distinguishing between speakers. Additionally, background noise was notably prominent across all sound categories.

Results of Filter Design #2

Filter Design #2 introduces a 50% overlap of subbands. This means that each band overlaps with its two adjacent bands by 25%, to minimize frequency gaps and sharp changes at the band boundaries.

As shown in Table 2, the filter demonstrated relatively good performance, with an overall score of 36.81. The filter performed well for speech and man-made sounds, with scores of 35.22 and 33.40 respectively. This design yielded the best results for environmental sounds, with a score of 53.44, which can be attributed to the sound naturalness metric (Appendix).

In regard to other specific evaluation metrics, the filter resulted in particularly high ratings for speech clarity (42.34) and word intelligibility (42.19), as shown in the Appendix. These metrics depend on the proper representation of frequency components, such as consonants, that are critical for distinguishing words. By implementing overlapping subbands, all frequency information is preserved, which also reduces ringing that would otherwise occur when the signal is split at non-overlapping bands [1] [2].

Results of Filter Design #3

In Filter Design #3—an iteration of Filter Design #1, the Butterworth filters are replaced with Chebyshev filters.

Overall, this filter design showcases a relatively low performance, with an average score of 23.68, as shown in Table 2. This design performs the weakest in the speech category (21.43), followed by man-made sounds (22.48) and environmental sounds (36.25). The lower speech scores suggest that there is greater distortion in the processed speech output.

Compared to Filter Design #1, the switch to a Chebyshev filter appears to degrade the quality of processed sounds. Chebyshev filters often introduce ripples in the passband to prioritize steep roll-offs over signal smoothness [3]. These ripples can distort speech signals by affecting the clarity of critical frequencies, thereby reducing word intelligibility in processed sounds. However, it has a comparatively moderate score in environmental sounds, which suggests that the Chebyshev filter handles complex soundscapes better than speech.

Results of Filter Design #4

For Filter Design #4, the Butterworth filters are replaced with Elliptic filters.

This filter design achieves a slightly higher average score than Filter Design #3. However, it still performs relatively poorly, with an average score of 24.38, as shown in Table 2. This design performs the weakest in the speech category (20.07), followed by man-made sounds (25.33) and environmental sounds (38.75). The lower scores for speech suggest that there is greater distortion in the processed speech output.

Elliptic filters often have sharp roll-offs and equalized ripple characteristics in both the passband and stopband [4]. While these properties help ensure precise frequency separations, they sacrifice smooth transitions. As a result, these ripples disrupt the signal which directly affects speech clarity. However, the stronger performance in environmental sounds may suggest that the steep attenuations in Elliptic filters help isolate noise components more effectively, which may be less critical for speech clarity.

Results of Filter Design #5

Filter Design #5 replaced the logarithmic spacing of sub-bands with linear spacing, distributing the frequency bands evenly across the range of 100 Hz to 8 kHz.

This filter design resulted in the highest overall score among all iterations, achieving 42.72. It performed especially well in the speech and man-made sound categories, with scores of 40.52 and 44.79 respectively. Testing revealed that this filter excelled in word intelligibility and noise reduction, as shown in Appendix. However, this came at the cost of naturalness. Listeners described the processed sounds as monotonic and robotic, lacking the tonal variation and intonation of natural speech.

This outcome aligns with the understanding that linear spacing does not align with the logarithmic nature of the human auditory system [5]. The human ear is more sensitive to changes at lower frequencies, making logarithmic spacing more effective at preserving expressive tones in speech [5].

Effects of Changing the Number of Channels

From the preliminary evaluations, the filter designs that performed the best for speech were selected, as it is the most critical aspect of everyday applications of the cochlear implant. As a result, Filter Designs #2 (overlapping subbands) and #5 (linear spacing of bands) were chosen as they achieved the highest scores overall and in the speech category. In contrast, Filter Designs #1, #3, and #4 performed well for environmental sounds but were less effective at processing speech. Therefore, adjustments to the number of channels were applied to Filter Designs #2 and #5.

Increasing the number of channels allows for finer frequency separation, which enhances sound clarity and improves noise reduction by isolating specific frequency components more effectively. This adjustment particularly benefits Filter Design #2, as it addresses its lowest-scoring metrics—multi-speaker scenarios and noise reduction—leading to significant potential improvements.

However, for Filter Design #5, which demonstrated weak performance in the speech clarity metric (score of 36.25, see Appendix), increasing the number of channels did not resolve this issue. Instead, it increased the unnatural, robotic quality of speech sounds. Moreover, it became more difficult to recognize individual speakers. This means that with linear band spacing, increasing the number of channels may compromise word intelligibility and naturalness.

It is therefore concluded that the most optimal cochlear implant is based on Filter Design #2, with an increased number of channels to address its previously weaker performance in multi-speaker scenarios and noise reduction.

Optimal Number of Channels

Increasing the number of bands also increases computational complexity, which requires more hardware to meet these processing requirements. For the cochlear implant, the physical bulkiness of the implant must be minimized for user comfort, and the production cost must also be reduced for large-scale manufacturing.

Typically, popular cochlear implants, such as those developed by Cochlear Limited, use 22 frequency channels [6] [7]. With other designs supporting 12 channels, this deviation of 10 can be further extrapolated, to establish an upper bound of 32 channels [7]. According to a study focused on speech sound processing, there is no statistically significant difference in performance beyond an increase of 8 bands [8]. However, it was experimentally determined that changing the number of bands to 32 still resulted in noticeable improvements to multi-speaker performance and noise reduction.

In consideration of the upper bound of 32, and the priority of speech sound processing, the final design of Filter Design #2 (overlapping subbands) with 32 channels is selected. This ensures optimized benefits for noise reduction and multi-speaker performance while staying within the upper bound to balance cost and hardware requirements for the implant.

Bonus Task

For the top-performing filter design, the following quantitative evaluation metric is proposed.

Pitch Contour Fidelity— *Speech and Man-Made Sounds (Musical Instruments)*

Pitch contour fidelity evaluates how accurately the processed speech preserves the natural variations in pitch found in the original audio. In speech, pitch contours reflect variations in intonation, stress, and rhythm that help speakers convey emotion and meaning (i.e. questions versus statements) [9]. For music, pitch contours represent melodic lines, tonal variations, and expressive qualities like vibrato and glissando [9]. By maintaining pitch contour fidelity, it ensures that the processed output audio retains its intended emotional, semantic, and tonal characteristics [9].

Thus, this metric is proposed to assess deviations in pitch contour to determine the level of pitch distortion introduced during the filter processing.

In MATLAB, the following tools will be used to measure this metric [10] [11]:

- pitch function: Extracts pitch contours over time, allowing direct comparison between original and processed signals.
- harmonicRatio function: Assesses the tonal quality, which contributes to the fidelity of musical pitch variations

Scoring System:
A scoring system from 0% to 100% will be used to evaluate how well the original pitch contours are preserved

- 0% (Undesirable): Significant distortions or complete loss of pitch variation.
- 50% (Moderate): Some pitch contours are preserved, but essential nuances are lost.
- 100% (Excellent): Near-perfect preservation of pitch contours, maintaining natural expressiveness and tonal accuracy.

Key Parameters:

1. Intonation Patterns: Variations in pitch and stress across different speech types (e.g., declarative, interrogative, or emotional tones)
2. Pitch Range: Evaluation across low, mid, and high frequencies in speech and music.
3. Melodic Fidelity: Accuracy in retaining musical phrasing, vibrato, and dynamic pitch transitions.
4. Robustness: Evaluation under varying noise levels and overlapping acoustic scenarios.

Conclusion

In conclusion, by varying parameters such as subband overlap, filter type, and spacing, the evaluations reveal significant differences in cochlear implant performance. Filter Design #2, which has a 50% overlap of subbands, emerged as the best choice, providing an effective filter design for speech processing needs. Further refinement, including increasing the number of channels to 32, optimizes noise reduction and performance in multi-speaker scenarios, ensuring that the design meets both auditory clarity and hardware feasibility.

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Appendix

Sound samples were scored based on evaluation metrics identified in Phase 1. The scoring process is documented in the following file:

Sound Evaluations for Filter Designs